

Data Flow Diagram & User Stories - ShopEZ Stock Trading Platform

Parameter	Details
Date	18 June 2026
Team ID	SE-09
Project Name	ShopEZ Stock Trading Platform
Maximum Marks	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 (Context Diagram) Description

At Level 0 (Context Diagram), the ShopEZ Stock Trading Platform is represented as a single central process. Three external entities interact with it:

- **Trader (User):** Submits registration info, logs in, searches stocks, submits buy/sell trade orders, views transaction history, and views portfolio balance.
- **Live Stock Feed (External API):** Provides real-time stock price quotes (simulated tick-by-tick or fetched via API).
- **Admin:** Views metrics counts (total users, total trades, total stocks), adds/deletes stock tickers, and moderates trader accounts.

DFD Level 1 Description

- **Process 1.0 (Auth):** Manages user registrations, encrypts credentials using bcrypt hashing, and handles session security via JWT token signatures.
- **Process 2.0 (Price Simulation):** Background task that updates simulated stock prices in the database at regular intervals to mimic live market fluctuations.
- **Process 3.0 (Trade Engine):** Validates buy/sell trade requests against user balance and stock holdings, registers transaction records, and updates trader portfolios.
- **Process 4.0 (Portfolio Aggregation):** Aggregates active user holdings, computes average purchase costs, and calculates real-time profit/loss metrics.
- **Process 5.0 (Admin Panel):** Aggregates collection counts (total users, total trades) and handles stock ticker catalog updates.

- **Data Stores:** Users Collection, Stocks Collection, Transactions Collection, Portfolios Collection.
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User Stories

Use the below template to list all the user stories for the product.